

2 MARKS

UNIT I – INTRODUCTION TO BUSINESS ANALYTICS

1. What is Business Analytics? Explain with an example.

Business Analytics is the process of collecting, analyzing, and interpreting data to support business decision making. It helps organizations improve performance and solve business problems.

Example:

An online shopping company analyzes customer purchase history to recommend products.

2. Define Data Science.

Data Science is a field that uses scientific methods, algorithms, and analytical techniques to extract useful information from data.

It combines:

- Statistics
 - Programming
 - Machine Learning
 - Data Analysis
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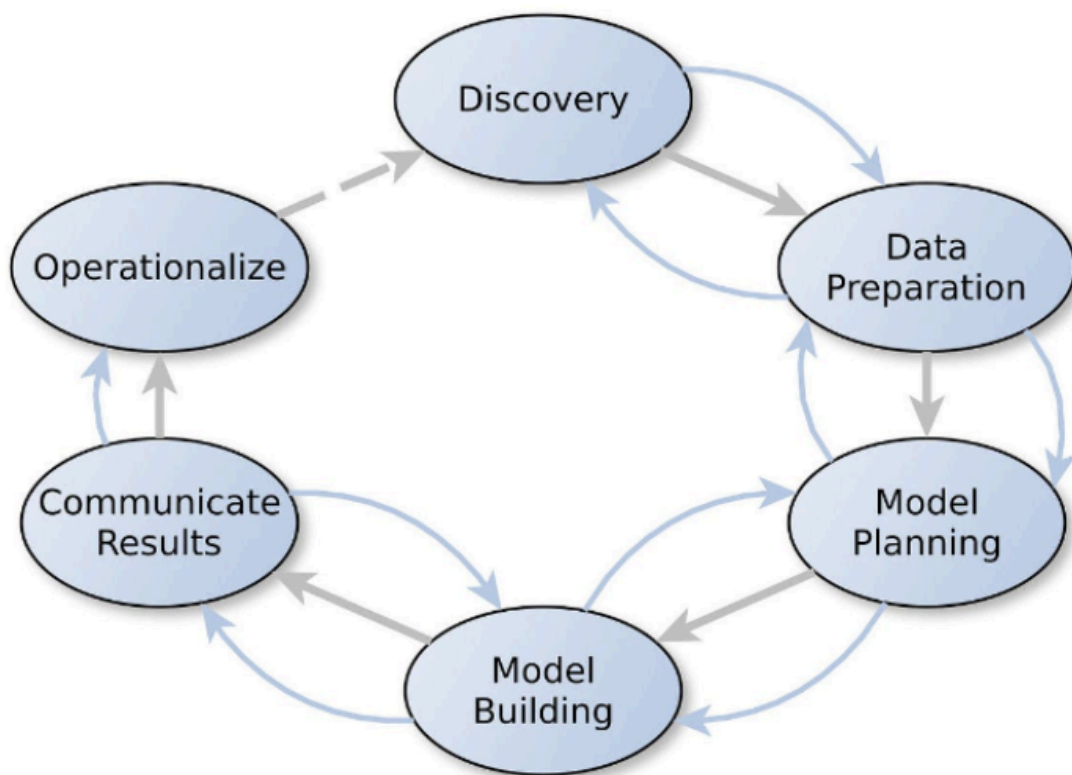
3. List and explain the types of Analytics.

The main types of analytics are:

1. **Descriptive Analytics** – Explains what happened in the past.
 2. **Diagnostic Analytics** – Explains why it happened.
 3. **Predictive Analytics** – Predicts future events.
 4. **Prescriptive Analytics** – Suggests suitable actions.
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4. Explain the term Analytics Life Cycle.

Analytics Life Cycle is a step-by-step process followed to solve business problems using data analysis techniques.



5. What is Business Problem Definition?

Business Problem Definition is the process of identifying and clearly understanding a business issue before starting analysis.

It helps:

- Define objectives
- Reduce confusion
- Find suitable solutions

Example:

A company identifying “decrease in sales” as a business problem.

6. Define Data Collection. Mention its sources.

Data Collection is the process of gathering data from different sources for analysis.

Sources of Data:

1. Surveys
2. Databases
3. Websites
4. Social Media

Example:

Collecting customer feedback through forms.

7. Explain Data Preparation.

Data Preparation is the process of cleaning and organizing raw data into a proper format for analysis.

Activities involved:

- Removing duplicate data
 - Correcting errors
 - Handling missing values
 - Formatting data
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8. What is Hypothesis Generation?

Hypothesis Generation is the process of creating assumptions or possible solutions for a business problem.

9. Define Modeling in Business Analytics.

Modeling is the process of applying statistical or mathematical methods to analyze data and identify patterns.

Purpose:

- Predict future trends
- Support decision making
- Solve business problems

Example:

Using a sales forecasting model to predict future revenue.

10. What is Validation and Evaluation?

Validation and Evaluation are processes used to test the accuracy and effectiveness of analytical models. It improves model performance and reduces errors.

11. Explain Deployment in Analytics.

Deployment is the process of implementing an analytical model into a real business environment.

Example: A bank using a fraud detection model in daily transactions.

12. State the advantages of Business Analytics.

Advantages of Business Analytics:

1. Improves decision making
 2. Reduces business risk
 3. Increases productivity
 4. Helps identify market trends
 5. Improves customer satisfaction
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13. Mention the applications of Business Analytics.

- Marketing analysis
 - Financial analysis
 - Healthcare management
 - Supply chain management
 - Human resource management
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14. Differentiate between Data Science and Business Analytics.

Data Science	Business Analytics
Focuses on extracting knowledge from data	Focuses on solving business problems
Uses AI and Machine Learning	Uses statistical analysis
Handles structured and unstructured data	Mainly handles structured data
More technical	More business-oriented

15. Differentiate between Descriptive Analytics and Predictive Analytics.

Descriptive Analytics	Predictive Analytics
Explains past events	Predicts future events
Uses historical data	Uses statistical models
Gives reports and summaries	Gives future forecasts
Simple analysis	Advanced analysis

UNIT II – BUSINESS INTELLIGENCE

1. Define Business Intelligence (BI).

Business Intelligence is a technology-driven process used to collect, analyze, and present business data for better decision making. Used for Report generation

2. What is a Data Warehouse?

A Data Warehouse is a centralized storage system used to store large amounts of historical data collected from multiple sources.



3. Define Data Mart.

A Data Mart is a smaller part of a data warehouse designed for a specific department such as sales or finance.

Example: A sales department maintaining only customer and sales data.

4. List the types of decisions in Business Intelligence.

The types of decisions are:

1. **Strategic Decisions** – Long-term decisions taken by top management.
 2. **Tactical Decisions** – Medium-term decisions taken by middle management.
 3. **Operational Decisions** – Daily routine decisions.
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5. Explain the Decision Making Process.

Decision Making Process is the method of selecting the best solution among different alternatives.

Steps:

1. Identify the problem
 2. Collect information
 3. Generate alternatives
 4. Select the best solution
 5. Implement decision
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6. Define Decision Support System (DSS).

Decision Support System is a computer-based system that helps managers make effective decisions using data and models. Used for faster decision making.

7. What is OLAP?

OLAP (Online Analytical Processing) is a technology used for multidimensional analysis of business data.

Uses:

- Sales analysis
 - Financial reporting
 - Trend analysis
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8. Define Knowledge Management.

Knowledge Management is the process of creating, storing, sharing, and utilizing knowledge within an organization.

9. Explain the importance of Decision Support Systems.

1. Helps managers make better decisions
 2. Improves problem-solving ability
 3. Saves time in analysis
 4. Supports business planning
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10. State the role of Business Intelligence in organizations.

Role of Business Intelligence:

- Supports decision making
- Improves business performance
- Identifies market trends
- Enhances operational efficiency

11. Mention the applications of Business Intelligence.

Applications of BI:

1. Customer relationship management
 2. Financial analysis
 3. Sales forecasting
 4. Inventory management
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12. State the advantages of Data Warehouse.

Advantages of Data Warehouse:

1. Centralized data storage
 2. Faster report generation
 3. Better data consistency
 4. Improved business analysis
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13. Differentiate between Data Warehouse and Data Mart.

Data Warehouse	Data Mart
Stores organization-wide data	Stores department-specific data
Large in size	Small in size
More complex	Less complex
Higher cost	Lower cost

14. Differentiate between OLTP and OLAP.

OLTP	OLAP
Used for transaction processing	Used for analytical processing
Handles current data	Handles historical data
Supports daily operations	Supports decision making
Fast insert/update	Fast query analysis

UNIT III – BUSINESS FORECASTING

1. Define Business Forecasting.

Business Forecasting is the process of predicting future business conditions using historical data, statistical techniques, and market trends.

Example: A company predicting next year's product sales based on previous sales records.

2. What is Predictive Analytics?

Predictive Analytics is the process of using data, statistical methods, and machine learning techniques to predict future outcomes.

Example: Netflix recommends movies based on user watch history.

3. Explain the types of Business Forecasting.

Qualitative Forecasting	Quantitative Forecasting
Based on opinions	Based on numerical data
Suitable when data is limited	Suitable when historical data is available
Subjective method	Objective method

4. Define Data Mining.

Data Mining is the process of extracting useful patterns, relationships, and information from large datasets.

5. Explain Logic Driven Models.

Logic Driven Models are forecasting models that use predefined rules, assumptions, and logical relationships for decision making.

Example: “If temperature increases in the summer season, ice cream sales will increase.”

6. Explain Data Driven Models.

Data Driven Models use historical data and statistical analysis to identify patterns and make predictions.

Example: Predicting stock market trends using previous market data.

7. Define Machine Learning.

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and improve automatically without explicit programming.

8. State the role of Predictive Analytics in Business.

Role of Predictive Analytics:

1. Predicts future trends
 2. Improves business planning
 3. Reduces operational risks
 4. Increases customer satisfaction
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9. Mention the applications of Predictive Analytics.

Applications of Predictive Analytics:

1. Weather forecasting
 2. Fraud detection
 3. Sales forecasting
 4. Healthcare prediction
 5. Customer behavior analysis
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10. State the advantages of Data Mining.

Advantages of Data Mining:

1. Identifies hidden patterns
 2. Improves decision making
 3. Reduces business risk
 4. Saves time and cost
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12. Differentiate between Forecasting and Prediction.

Forecasting	Prediction
Estimates future trends	Estimates future outcomes
Based on historical data	Based on assumptions and models
Used in business planning	Used in data analysis
Long-term focus	Short-term focus

13. Differentiate between Data Mining and Machine Learning.

Data Mining	Machine Learning
Extracts patterns from data	Learns automatically from data
Used for analysis	Used for prediction
Requires human guidance	Supports automation
Data-oriented process	Algorithm-oriented process

14. Explain the importance of Business Forecasting.

Importance of Business Forecasting:

1. Helps in future planning
 2. Reduces uncertainty
 3. Improves resource management
 4. Supports business growth
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15. Mention the applications of Machine Learning.

Applications of Machine Learning:

1. Chatbots
2. Recommendation systems
3. Self-driving cars
4. Spam filtering
5. Medical diagnosis

UNIT IV – HR & SUPPLY CHAIN ANALYTICS

1. Define HR Analytics.

HR Analytics is the process of collecting and analyzing employee-related data to improve human resource decisions.

Uses: Employee performance analysis

2. What is Human Resource Planning?

Human Resource Planning is the process of identifying the number and type of employees required for an organization.

Objectives:

- Right employee
 - Right job
 - Right time
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3. Define Recruitment.

Recruitment is the process of searching, selecting, and hiring suitable candidates for job positions.

Sources of Recruitment:

1. Internal recruitment
2. External recruitment

4. Explain Training and Development.

Training and Development improve employees' skills, knowledge, and performance.

Training	Development
Improves current job skills	Improves future abilities
Short-term process	Long-term process
Job-oriented	Career-oriented

5. What is a Supply Chain Network?

A Supply Chain Network is a system connecting suppliers, manufacturers, warehouses, distributors, and customers.

Flow of Supply Chain

Supplier → Manufacturer → Warehouse → Customer

6. Explain Demand Planning.

Demand Planning is the process of forecasting customer demand to ensure product availability.

Benefits:

- Improves inventory management
- Increases customer satisfaction

7. Define Inventory Planning.

Inventory Planning is the process of maintaining the correct quantity of stock to meet customer demand.

8. What is Logistics?

Logistics is the process of transporting, storing, and managing goods efficiently from source to customer.

Activities involved:

- Transportation
 - Warehousing
 - Packaging
 - Distribution
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9. Mention the applications of HR Analytics.

Applications of HR Analytics:

1. Recruitment analysis
 2. Employee performance evaluation
 3. Attendance tracking
 4. Employee retention analysis
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10. Mention the applications of Supply Chain Analytics.

Applications of Supply Chain Analytics:

1. Inventory management
 2. Transportation planning
 3. Supplier analysis
 4. Demand forecasting
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11. State the advantages of HR Analytics.

Advantages of HR Analytics:

1. Improves hiring decisions
 2. Increases employee productivity
 3. Reduces employee turnover
 4. Supports workforce planning
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12. State the advantages of Supply Chain Analytics.

Advantages of Supply Chain Analytics:

1. Reduces operational cost
 2. Improves delivery performance
 3. Enhances inventory control
 4. Improves customer satisfaction
-

13. Differentiate between HR Analytics and Supply Chain Analytics.

HR Analytics	Supply Chain Analytics
Focuses on employees	Focuses on product movement
Used in HR department	Used in logistics department
Improves workforce management	Improves supply chain efficiency
Analyzes employee data	Analyzes inventory and transport data

14. Explain how HR Analytics helps predict employee demand.

HR Analytics analyzes historical employee data, workload, and business growth to predict future employee requirements.

Example: A retail company predicting the number of employees required during festival seasons.

15. Supply Management and Inventory Management

Supply Management	Inventory Management
Managing flow of goods from suppliers	Managing stock inside company
Procurement and supply	Storage and control of stock
Ensure supply availability	Maintain optimal inventory

UNIT V – MARKETING & SALES ANALYTICS

1. Define Marketing Strategy.

Marketing Strategy is a plan used by organizations to promote products and achieve business goals.

Objectives:

- Increase sales
 - Attract customers
 - Improve brand value
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2. Explain the Marketing Mix.

Marketing Mix refers to the 4Ps of marketing:

1. **Product** – Goods or services offered
 2. **Price** – Amount charged
 3. **Place** – Distribution method
 4. **Promotion** – Advertising and sales activities
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3. Define Customer Behaviour.

Customer Behaviour is the study of how customers select, purchase, and use products or services.

4. What is Sales Planning?

Sales Planning is the process of setting sales targets and preparing strategies to achieve them.

5. Explain the Selling Process.

Selling Process is a sequence of steps followed to sell products or services successfully.

Steps:

1. Prospecting
 2. Approach
 3. Presentation
 4. Handling objections
 5. Closing the sale
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6. Mention the applications of Marketing Analytics.

Applications of Marketing Analytics:

1. Customer segmentation
2. Campaign analysis
3. Product recommendation
4. Market trend analysis

7. Mention the applications of Sales Analytics.

Applications of Sales Analytics:

1. Sales forecasting
 2. Revenue analysis
 3. Sales performance tracking
 4. Customer analysis
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8. State the advantages of Marketing Analytics.

Advantages of Marketing Analytics:

1. Improves marketing strategies
 2. Understands customer needs
 3. Increases customer satisfaction
 4. Improves campaign performance
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9. State the advantages of Sales Analytics.

Advantages of Sales Analytics:

1. Improves sales forecasting
 2. Tracks sales performance
 3. Identifies sales trends
 4. Increases revenue
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10. Explain Predictive Analytics in Marketing.

Predictive Analytics in Marketing uses customer data and statistical techniques to predict customer behavior and future purchasing trends.

Example: Amazon recommends products based on previous purchases.

11. Define Customer Segmentation.

Customer Segmentation is the process of dividing customers into groups based on common characteristics such as age, income, or interests.

12. Differentiate between Marketing and Sales.

Marketing	Sales
Creates customer awareness	Converts leads into customers
Focuses on customer needs	Focuses on revenue generation
Long-term process	Short-term process
Includes promotion activities	Includes direct selling

13. Differentiate between Marketing Analytics and Sales Analytics.

Marketing Analytics	Sales Analytics
Focuses on marketing performance	Focuses on sales performance
Analyzes customer behavior	Analyzes sales data
Improves marketing campaigns	Improves sales strategies
Used by marketing team	Used by sales team

14. Common software for sales analysis

- Microsoft Excel
 - Power BI
 - Tableau
 - Salesforce
 - Google Analytics
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15. What are 7R's?

7P's is an extended marketing mix:

1. Right Product.
2. Right Quality
3. Right Condition
4. Right Place
5. Right Time
6. Right Customer(people)
7. Right Price